

Homework 6

due 22.11.2023, 14:00

Exercise 6.1 - Matrix Multiplication (10 points)

Write a program that reads two matrices from files, multiplies them in parallel and prints the result to a file again. Implement the algorithm from the lecture, that relies on an “Allgather” of a matrix column. The first row of a matrix file contains m and n , the number of rows and columns of a matrix A . The remaining lines contain one matrix element each, in the order $a_{00}, a_{01}, \dots, a_{0n-1}, a_{10}, \dots, a_{m-1n-1}$.

- Process 0 should be responsible for all I/O.
- The matrices can be non-square. For a multiplication $\underbrace{C}_{m \times k} = \underbrace{A}_{m \times n} * \underbrace{B}_{n \times k}$ every process should dynamically allocate space for local matrices of size $\bar{m} \times k$, $\bar{m} \times n$ and $\bar{n} \times k$, with $\bar{m} = m/\text{nproc}$ and $\bar{n} = n/\text{nproc}$. The easiest is to use 1D arrays, such that e.g. local matrix element a_{ij} is accessed via `a[j+n*i]`.
- To test your program, use the two matrices `A.txt` and `B.txt` that you can find in `/work/korzec/LAB2/ex6/` on `stromboli` (avoid making copies, these files can be accessed from the compute nodes for reading). Your program should create the output `C.txt`, which can be visualized as an image with the matlab script `check.m`. Include this image in your PDF.
- For the test a good number of processes is `nproc=10` or `nproc=30`.